

Supplementary Material

(For Online Publication Only)

Appendix A

Tables and Figures

Table A.1 – Effect of alcohol ban on completed years of schooling, enrollment (current) and dropout status of children in interior Bihar

	Completed Years of Schooling	Enrollment Status	Dropout Status
	(1)	(2)	(3)
Bihar Interior × Post	0.058 (0.035)	-0.013** (0.006)	0.003 (0.003)
District FE	Yes	Yes	Yes
Survey Year FE	Yes	Yes	Yes
Birth Cohort FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Observations	216,969	263,710	255,770
R-squared	0.020	0.093	0.061
Mean	0	0.943	0.028

Notes: The dependent variables are the completed years of schooling (child's grade standardized with respect to the mean and standard deviation of age-specific grade), enrollment (dummy takes value 1 if the child is currently enrolled and 0 otherwise) and dropout status (dummy takes value 1 if the child is a dropout and 0 otherwise); the sample is restricted to children from Jharkhand and the interior districts of Bihar; individual and household level controls include child's age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.2 – Effect of alcohol ban on teachers' attendance rate in interior Bihar

	Teachers' Attendance Rate			
	(1)	(2)	(3)	(4)
Bihar Interior × Post	-0.061** (0.025)	-0.060** (0.025)	-0.062** (0.026)	-0.077*** (0.027)
District FE	No	Yes	Yes	Yes
Survey Year FE	No	No	Yes	Yes
Controls	No	No	No	Yes
Observations	7,688	7,688	7,688	6,784
R-Squared	0.020	0.033	0.057	0.073
Mean	0.818	0.818	0.818	0.819

Notes: The dependent variable is the ratio of different types of teachers present in the school on the day of survey to the total number of teachers appointed; the sample is restricted to schools from Jharkhand and the interior districts of Bihar; school level controls include number of grades offered, types of posts offered in school (principal, regular and para teachers), type of school, provision of mid-day-meal, presence of boundary wall, tap water in the school and location, blackboard and joint status of class 2 (controls are limited due to missing values); data source is the ASER school surveys from 2009-14, 2016 and 2018; standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.3 – Effect of alcohol ban on student attendance rate in interior Bihar

	Students' Attendance Rate			
	(1)	(2)	(3)	(4)
Bihar Interior $\times$ Post	-0.051** (0.019)	-0.050** (0.019)	-0.052*** (0.019)	-0.069*** (0.018)
District FE	No	Yes	Yes	Yes
Survey Year FE	No	No	Yes	Yes
Controls	No	No	No	Yes
Observations	7,941	7,941	7,941	7,050
R-Squared	0.021	0.103	0.116	0.156
Mean	0.573	0.573	0.573	0.573

Notes: The dependent variable is the ratio total students present on the day of survey to the total students enrolled in school; the sample is restricted to schools from Jharkhand and the interior districts of Bihar; school level controls include total school enrollment, number of grades offered, type of school, provision of mid-day-meal, presence of boundary wall, tap water in the school and location, blackboard and joint status of class 2 (controls are limited due to missing values); data source is the ASER school surveys from 2009-14, 2016 and 2018; standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4 – Heterogeneous effect of alcohol ban on reading and math scores by baseline alcohol consumption (alternative definition)

	Standardized			
	Reading Score		Math Score	
	Consumption		Consumption	
	High (1)	Low (2)	High (3)	Low (4)
Bihar Interior $\times$ Post	0.094*** (0.027)	0.052 (0.036)	0.129*** (0.027)	0.066 (0.040)
District FE	Yes	Yes	Yes	Yes
Survey Year FE	Yes	Yes	Yes	Yes
Birth Cohort FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	152,858	137,944	152,296	137,341
R-Squared	0.473	0.476	0.477	0.482

Notes: The dependent variables are reading and math scores, which have been standardized with respect to the control group mean and standard deviation; the coefficients can be interpreted as standard deviations; the sample is restricted to children from Jharkhand and the interior districts of Bihar; high consumption districts constitute all districts with baseline alcohol consumption above average consumption in the state and the opposite is true for low consumption districts; individual and household level controls include child's age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5 – Synthetic control method: control weights

Control Unit Name (State)	Control Unit Weight	
	Reading Score	Math Score
Uttar Pradesh	0	0
West Bengal	0.037	0.461
Jharkhand	0.907	0.439
Odisha	0.000	0.10
Chhattisgarh	0.000	0
Madhya Pradesh	0.056	0

Notes: The weights are computed in Stata using the *synth* package; covariates used to match include child's age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house and baseline test scores.

Table A.6 – Synthetic control method: covariate balance

	Reading Score		Math Score	
	Treated	Synthetic	Treated	Synthetic
Child Age	9.684	9.683	9.684	9.768
Child Gender (1=Boy)	0.539	0.532	0.539	0.520
School Class	4.017	4.113	4.017	4.222
Mother Literate (1=Yes)	0.375	0.357	0.375	0.488
Father Literate (1=Yes)	0.640	0.615	0.637	0.659
Household Size	7.384	6.692	7.384	6.292
Household Toilet	0.214	1.115	0.214	0.276
Household Type	1.888	1.465	1.889	1.568
Test Score (2009)	2.510	2.514	2.504	2.488
Test Score (2010)	2.570	2.560	2.531	2.495
Test Score (2011)	2.332	2.321	2.423	2.278
Test Score (2012)	2.271	2.223	2.16	2.15
Test Score (2013)	2.193	2.202	2.147	2.13
Test Score (2014)	2.197	2.197	2.128	2.124

Notes: The balance table is obtained from Stata using the *synth* package.

Table A.7 – The effect of alcohol ban on reading and math scores: Linear probability models

	(1)	(2)	(3)	(4)
<i>Panel A: A child can read</i>				
	(Letters)	(Words)	(Paragraph)	(Story)
Bihar Interior*Post	0.029*** (0.009)	0.039*** (0.013)	0.025** (0.012)	0.019* (0.011)
Observations	190,024	190,024	190,024	190,024
R-Squared	0.238	0.358	0.389	0.339
<i>Panel B: A child can recognize numbers and perform</i>				
	(0-9)	(10-99)	(Subtraction)	(Division)
Bihar Interior*Post	0.021** (0.009)	0.043*** (0.014)	0.041*** (0.012)	0.030*** (0.011)
Controls	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Survey Year FE	Yes	Yes	Yes	Yes
Birth Cohort FE	Yes	Yes	Yes	Yes
Observations	189,232	189,232	189,232	189,232
R-Squared	0.220	0.358	0.367	0.297

Notes: The dependent variables are the different levels of mastery from basic reading and arithmetic scores, all of which take a value 1 if the child has attained mastery in the particular foundational skill (as indicated by model labels) and 0 otherwise; the sample is restricted to children from Jharkhand and the interior districts of Bihar; individual and household level controls include child's age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure A.1: Yearly effect of alcohol ban on reading and math test scores of children in interior Bihar: sensitivity analysis

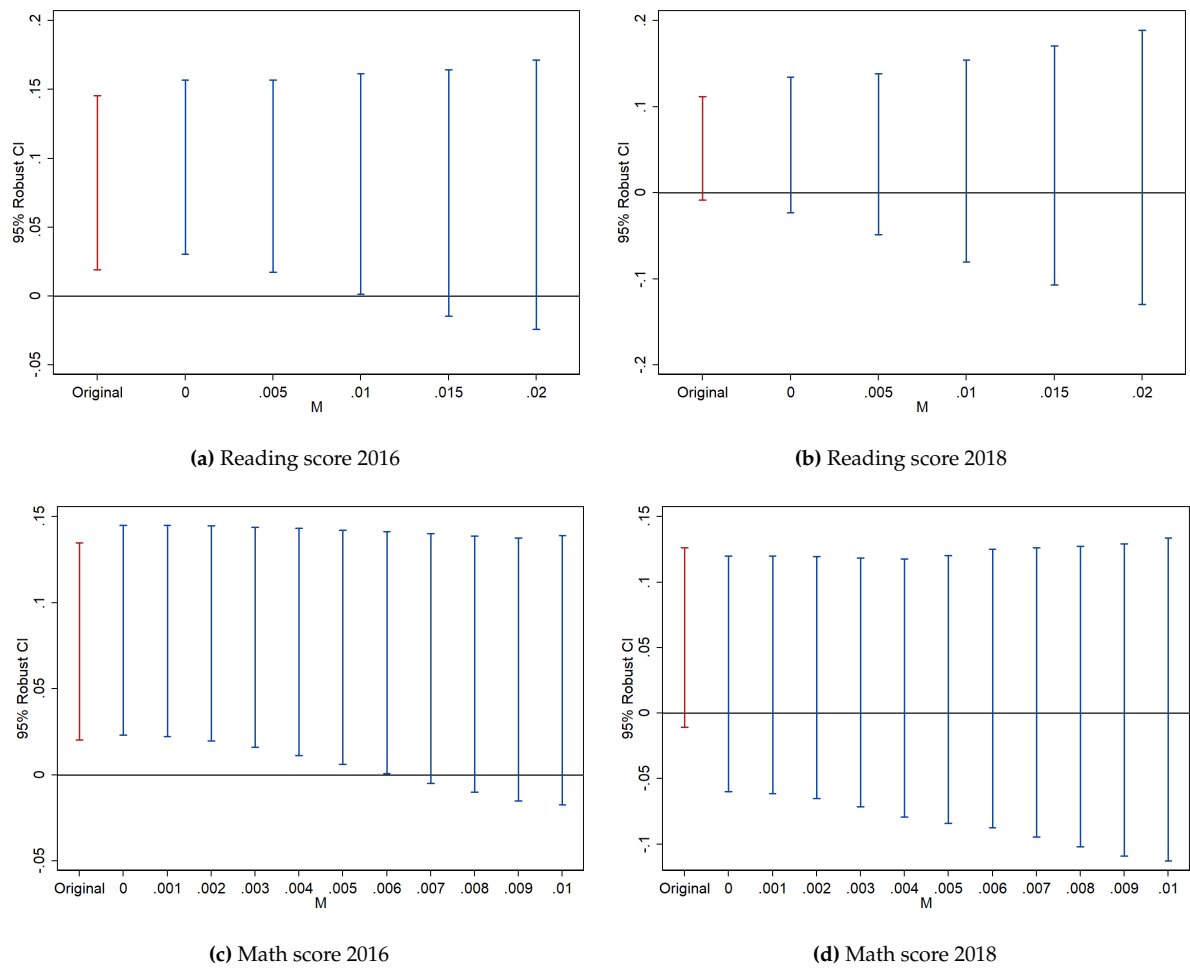
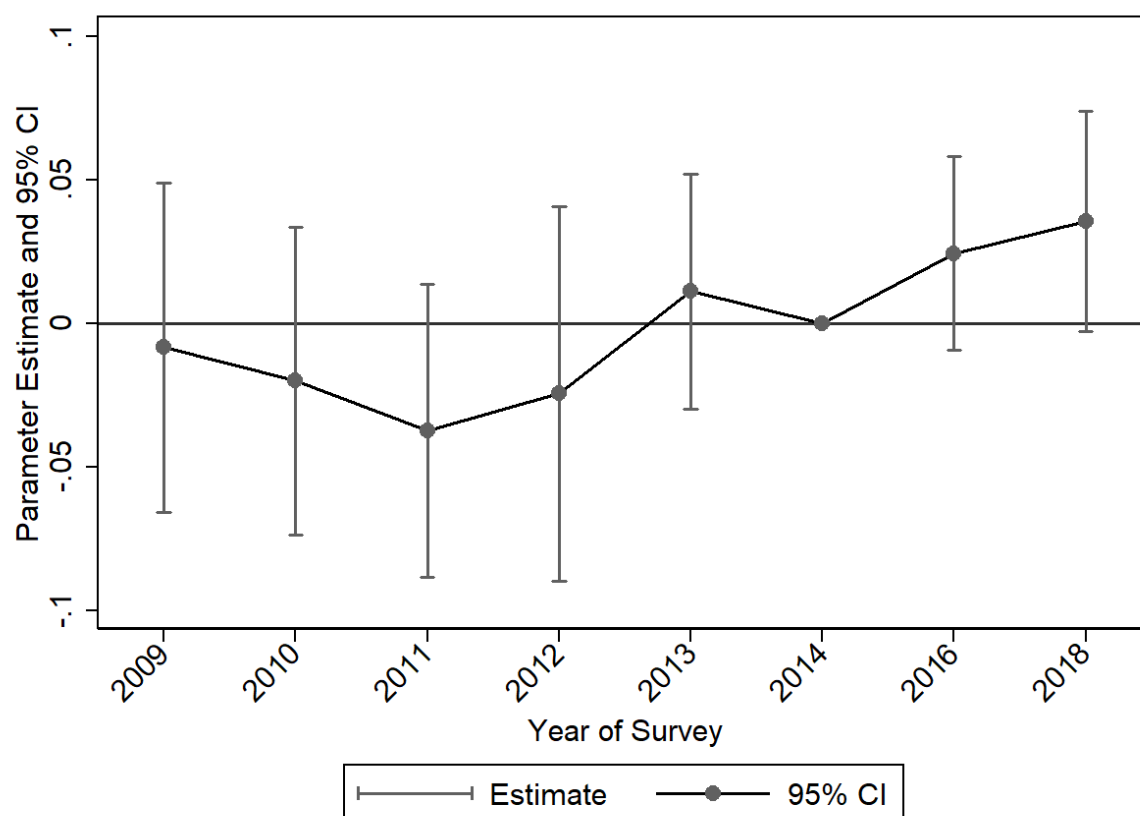


Figure A.2: Incidence of domestic violence in Bihar and Jharkhand over the years



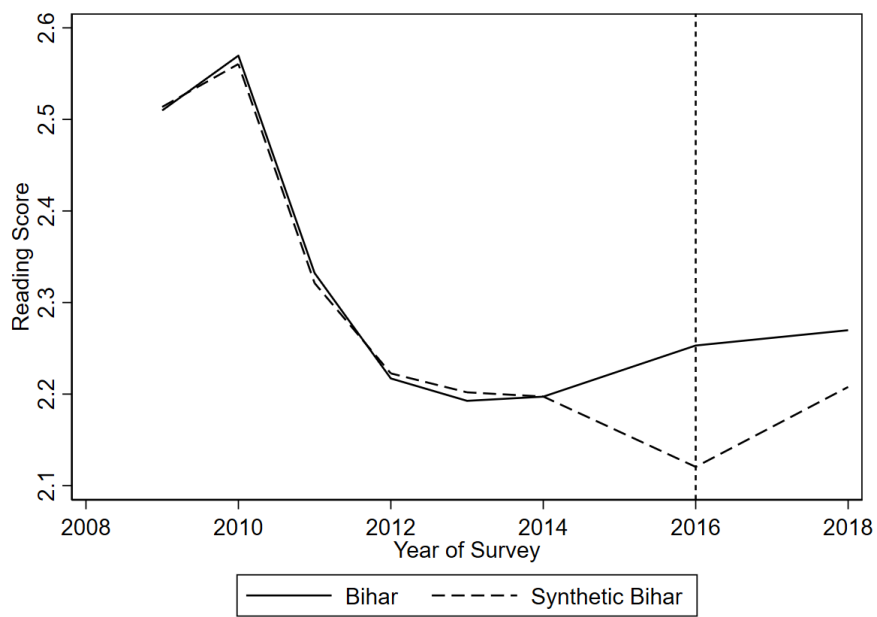
Notes: Data source is the annual report on Crimes in India, published by the National Crime Records Bureau (NCRB); The figure plots the number of cases registered under Section 498-A of the Indian Penal Code, which makes administering cruelty by the husband or his relatives on the wife an offense.

Figure A.3: Yearly effect of alcohol ban on after-class paid private tuition in interior Bihar: event study estimates

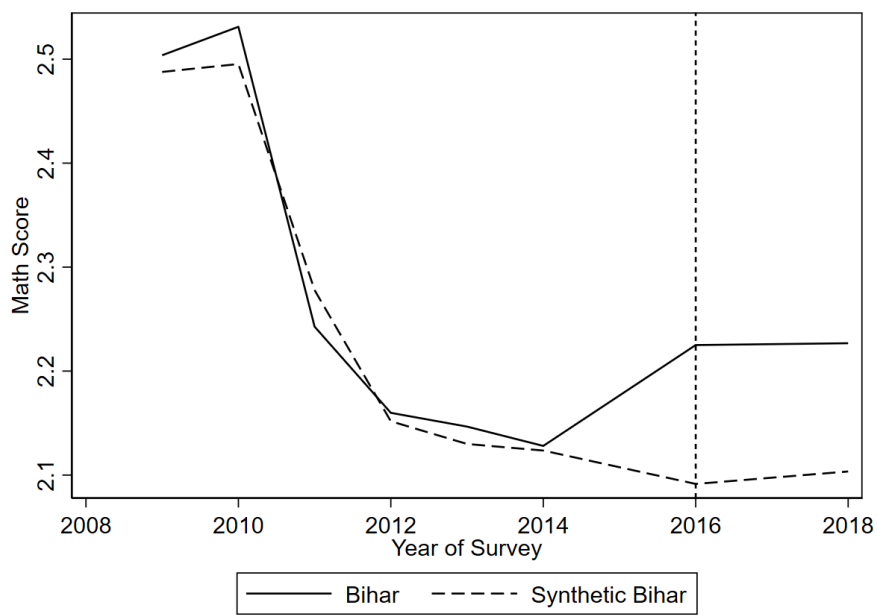


Notes: Wild clustered bootstrapped p-values for the two post periods, 2016 and 2018, are 0.157 and 0.077, respectively.

Figure A.4: Effect of alcohol ban on reading and math scores: outcome paths from the synthetic control method

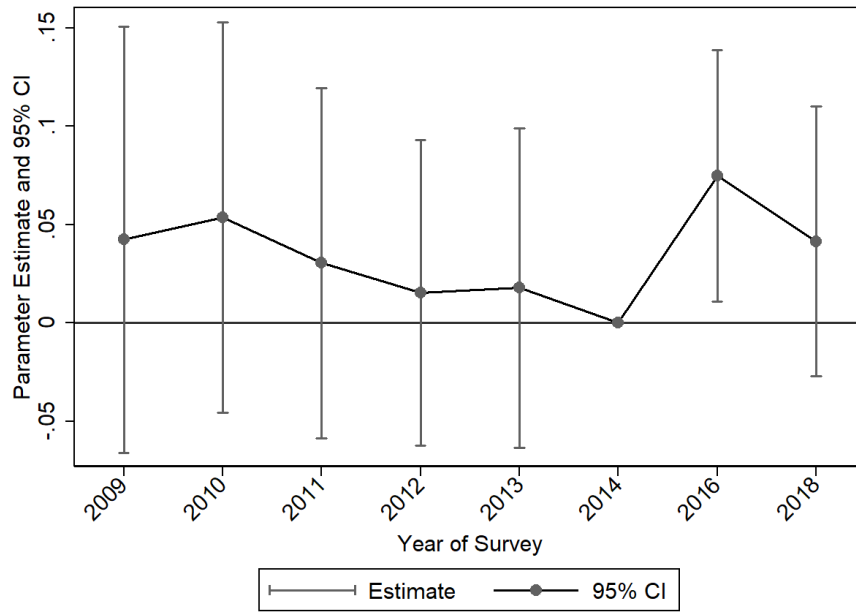


(a) Reading Score

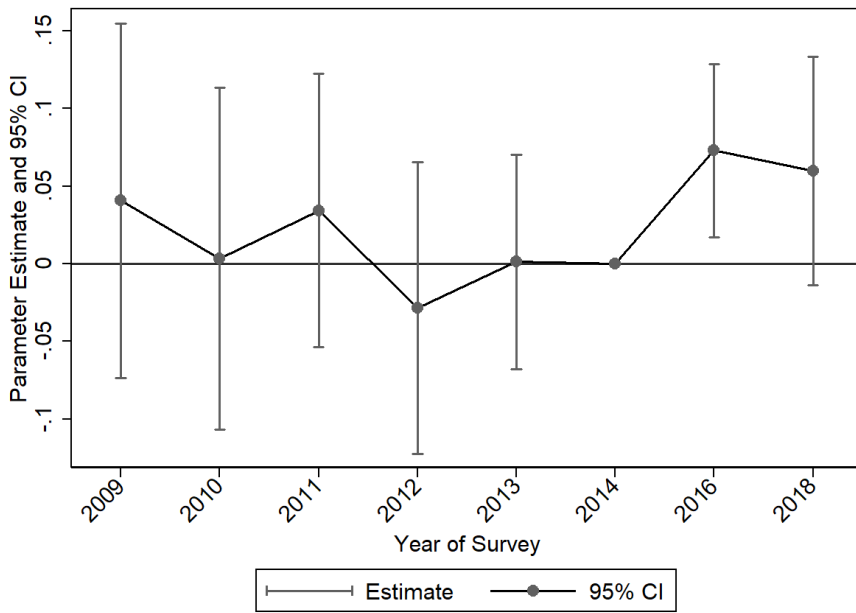


(b) Math Score

Figure A.5: Yearly effect of alcohol ban on reading and math test scores of children in Bihar (all districts): event study estimates



(a) Reading score



(b) Math score

Notes: Wild clustered bootstrapped p-values for the two post periods, 2016 and 2018, are 0.030 and 0.246 for reading and 0.013 and 0.126 for math scores, respectively.

B. Appendix B

Data Summary (NFHS)

To investigate the impact of state-wide alcohol ban on household environment in Bihar, we use data from the National Family Health Survey (NFHS) — a large multi-round survey conducted in a representative sample of households throughout India. Specifically, we use the latest two rounds of NFHS (IV and V) conducted in 2015-16 and 2019-2021, respectively. These two rounds are suitable for our analysis since they were conducted just before and after the implementation of the Bihar alcohol ban in 2016. For our purpose, we restrict the data to the interior districts of Bihar (the treatment group) and Jharkhand (the control group). Our analytical sample consists of ever-married women between the ages 18 and 49, who completed the domestic violence module of the survey. The domestic violence module captures the information related to intimate partner violence. Since the policy was enacted in April 2016, we only keep the observations recorded prior to this date in our sample from NFHS-4 (2015-16). Also, to ensure that our estimates are not impacted by COVID-19, we restrict our sample to women in Bihar who were interviewed prior to COVID-19.

We use the information on alcohol consumption by husbands/partners, their aggressive behavior (emotional, physical, sexual violence and controlling behavior by husband) and the bargaining power of women within the household. We also use various observable characteristics at the individual and household levels as controls in our regressions. For instance, at the individual level, we control for women's age, years of schooling, employment status, religion, caste, years since marriage, number of children aged 5 and less, husband's education, and place of residence. The household level controls include household size, gender and age of household head, and wealth index. To account for women's autonomy, we also controlled for whether she justifies wife beating on being unfaithful and whether she owns money that she can control. Table B.2 provides summary statistics of all the control variables both at the individual and household levels.

Outcome Variables

Our analysis includes a number of outcome variables. The first outcome variable is a dummy taking that takes the value 1 if the husband/partner consumes alcohol and '0' otherwise. For intimate partner violence, we construct three separate outcome variables measuring different forms of aggressive behavior by the husband. NFHS asks a list of questions corresponding to each of these aggressive behaviors listed in Table B.1. We code the responses to these questions as 1 if she ever faced such behavior and '0' otherwise. Using these responses, we construct three different indices measuring emotional violence, physical violence and sexual violence. To construct any particular index, we take average of all the related responses and standardize them using the mean and standard deviation of control group, so that each index has mean 0 and standard deviation 1 for control group. For example, to construct the emotional violence index, we take an average of the responses to three corresponding questions and then subtract the mean of the control group from the average score and divide it by the control

group's standard deviation. Additionally, we construct an aggregate index of intimate partner violence by taking the average of all 19 questions corresponding to different types of aggressive behavior by husband and standardize them using the mean and standard deviation of the control group.

Another outcome variable of interest is the husband's controlling behavior. We utilize information on the restrictions imposed by husbands on their wives to construct an index variable called "*controlling behavior by husband*". Following the methodology used for other indices, we take average of all the related responses and standardize them using the mean and standard deviation of the control group, so that each index has a mean of 0 and a standard deviation of 1 for the control group. We also construct an index for the bargaining power of women within the household. NFHS asks four questions listed in Table B.1 to assess women's intra-household bargaining power. We coded these responses as 2 if the woman is the sole decision-maker, 1 if both the woman and her husband are joint decision-makers and 0 if the decision is solely taken by the husband or by someone else. Again, we take an average of all these responses and standardize them using the control group's mean and standard deviation. Table B.2 provides summary statistics for all indexed outcome variables.

Tables

Table B.1 – NFHS questions for index construction

	Responses
<u><i>Emotional Violence</i></u>	
Husband ever humiliated you?	Ever Yes = 1
Husband ever threatened you with harm?	No = 0
Husband ever insulted or made you feel bad?	
<u><i>Physical Violence</i></u>	
Husband ever pushed, shook, or threw something at you?	Ever Yes = 1
Husband ever slapped you?	No = 0
Husband ever punched you with a fist or hit you with something harmful?	
Husband ever kicked or dragged you?	
Husband ever tried to strangle or burn you?	
Husband ever threatened you with a knife/gun or other weapon?	
Husband ever twisted your arm or pulled your hair?	
<u><i>Sexual Violence</i></u>	
Husband ever physically forced you into unwanted sex?	Ever Yes = 1
Husband ever forced you into other unwanted sexual acts?	No = 0
Husband ever forced you into other unwanted sexual acts?	
<u><i>Husband's Controlling Behavior</i></u>	
Husband is jealous if you talk with other men?	Yes = 1
Husband accuses you of unfaithfulness?	No = 0
Husband does not permit you to meet female friends?	
Husband tries to limit your contact with your family?	
Husband insists on knowing where you are?	
Husband doesn't trust you with money?	
<u><i>Bargaining Power</i></u>	
Who decides how your husband's earnings will be used?	Women herself = 2
Who usually makes decisions about making major household purchases?	Women & husband = 1
Who usually makes decisions about visits to your family or relatives?	Husband only = 0
Who usually makes decisions about visits to your family or relatives?	

Notes: These questions are taken from the domestic violence module of the NFHS women's questionnaire from NFHS-4 and NFHS-5.

Table B.2 – Summary statistics- NFHS

	Mean	SD	Min	Max	N
<i>Outcome Variables</i>					
Alcohol Consumption (1=Yes)	0.381	0.486	0	1	6,501
Emotional Violence Index	0.125	1.151	-0.321	4.512	6,501
Physical Violence Index	0.11	1.084	-0.563	4.273	6,501
Sexual Violence Index	0.062	1.094	-0.272	5.495	6,501
Overall Intimate Partner Violence Index	0.125	1.112	-0.536	5.498	6,501
Controlling Behavior Index	0.162	1.078	-0.982	2.788	6,447
Bargaining Power Index	-0.106	1.168	-2.486	3.092	6,136
<i>Women and household level characteristics</i>					
Age	31.743	8.160	18	49	6,351
Education (in years)	4.538	5.134	0	20	6,351
Husband's Education (in years)	6.548	5.281	0	20	6,351
Working Status (1=Yes)	0.198	0.398	0	1	6,351
<i>Religion</i>					
— Hindu	0.839	0.369	0	1	6,351
— Muslim	0.099	0.299	0	1	6,351
— Christian	0.019	0.136	0	1	6,351
— Sikh	0.001	0.028	0	1	6,351
— No Religion	0.0001	0.013	0	1	6,351
— Others	0.043	0.203	0	1	6,351
<i>Caste</i>					
— Scheduled Caste (SC)	0.189	0.392	0	1	6,351
— Scheduled Tribe(ST)	0.145	0.352	0	1	6,351
— Other Backward Caste (OBC)	0.531	0.499	0	1	6,351
— None of them	0.233	0.341	0	1	6,351
Years Since Marriage	15.39	8.998	0	44	6,351
Household Size	5.396	2.085	1	20	6,351
Household Head's Age	44.167	13.712	17	95	6,351
Household Head's Gender (1=Male)	0.813	0.39	0	1	6,351
<i>Wealth Index</i>					
— Poorest	0.455	0.498	0	1	6,351
— Poorer	0.239	0.427	0	1	6,351
— Middle	0.147	0.354	0	1	6,351
— Richer	0.095	0.293	0	1	6,351
— Richest	0.064	0.245	0	1	6,351
Children (age less than or equal to 5)	0.837	0.984	0	7	6,351
<i>Autonomy Indicators</i>					
— Unfaithful	0.192	0.394	0	1	6,351
— Owns Money	0.449	0.497	0	1	6,351

Notes: Authors calculations from NFHS household level data; the sample is restricted to women from Jharkhand and the interior districts of Bihar; data source is the NFHS-4 and NFHS-5.

C. Appendix C

Cost-Benefit Analysis: Present Discounted Value Calculation

In the post ban period 3,471,843 individuals of age 16-17 year will have a return of 0.68% on their original income of INR 24,064 for the next 30 years. Income in first period will increase by $24,064 \times 3,471,843 \times 0.0068$ equals to INR 560,373,742. In the second year the income will increase at the compound annual growth rate of 2% equalizes to an amount of $560,373,742 \times (1 + 0.02)$. In line with this, in 30th year, the incremental income would be $560,373,742 \times (1 + 0.02)^{29}$.

The present discounted value of increased income would be

$$= \frac{560,373,742}{(1 + 0.058)} + \frac{560,373,742(1 + 0.02)}{(1 + 0.058)^2} + \dots + \frac{560,373,742(1 + 0.02)^{29}}{(1 + 0.058)^{30}}$$

$$= \text{INR}9,636,757,010$$

D. Appendix D

Tables

Table D.1 – Effect of alcohol ban on the reading score of children in interior Bihar: alternate levels of clustering

	Standardized Reading Score							
	No Clusters	Robust Standard Errors	State Wild Clusters	State×Year Wild Clusters	State×Age Wild Clusters	District Wild Clusters	Age×Year Clusters	District×Year Clusters
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Bihar Interior × Post	0.076*** (0.009)	0.076*** (0.009)	0.076*** (0.001) [0.000]	0.076*** (0.010) [0.013]	0.076*** (0.016) [0.010]	0.076*** (0.028) [0.011]	0.076*** (0.012)	0.076*** (0.027)
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Survey Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Birth Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	190,024	190,024	190,024	190,024	190,024	190,024	190,024	190,024
R-squared	0.471	0.471	0.471	0.471	0.471	0.471	0.471	0.471
N Clusters	.	.	2	16	24	37	96	292

Notes: The dependent variable is the reading score, which has been standardized with respect to the control group mean and standard deviation; the coefficients can be interpreted as standard deviations; the sample is restricted to children from Jharkhand and the interior districts of Bihar; all regressions without district or year fixed effects include standalone treatment and post dummies but are not reported in the table; individual and household level controls include child's age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table D.2 – Effect of alcohol ban on the math score of children in interior Bihar: alternate levels of clustering

	Standardized Math Score							
	No Clusters	Robust Standard Errors	State Wild Clusters	State×Year Wild Clusters	State×Age Wild Clusters	District Wild Clusters	Age×Year Clusters	District×Year Clusters
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Bihar Interior × Post	0.102*** (0.009)	0.102*** (0.009)	0.102*** (0.001) [0.000]	0.102*** (0.014) [0.070]	0.102*** (0.024) [0.002]	0.102*** (0.029) [0.001]	0.102*** (0.019)	0.102*** (0.028)
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Survey Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Birth Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	189,232	189,232	189,232	189,232	189,232	189,232	189,232	189,232
R-squared	0.478	0.478	0.478	0.478	0.478	0.478	0.478	0.478
N Clusters	.	.	2	16	24	37	96	292

Notes: The dependent variable is the math score, which has been standardized with respect to the control group mean and standard deviation; the coefficients can be interpreted as standard deviations; the sample is restricted to children from Jharkhand and the interior districts of Bihar; all regressions without district or year fixed effects include standalone treatment and post dummies but are not reported in the table; individual and household level controls include child's age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table D.3 – Effect of alcohol ban on the after-class tuition of children in interior Bihar: alternate level of clustering

	After-class Private Tuition							
	No Clusters	Robust Standard Errors	State Wild Clusters	State×Year Wild Clusters	State×Age Wild Clusters	District Wild Clusters	Age×Year Clusters	District×Year Clusters
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Bihar Interior × Post	0.043*** (0.005)	0.043*** (0.005)	0.043*** (0.000) [0.000]	0.043*** (0.005) [0.046]	0.043*** (0.009) [0.001]	0.043*** (0.016) [0.015]	0.043*** (0.006)	0.043*** (0.015)
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Survey Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Birth Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	172,830	172,830	172,830	172,830	172,830	172,830	172,830	172,830
R-squared	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180
N Clusters	.	.	2	16	24	37	96	292
Mean	0.429	0.429	0.429	0.429	0.429	0.429	0.429	0.429

Notes: The dependent variable is after-class paid private tuition, a dummy that takes value 1 if the child goes to private tuition classes and 0 otherwise; the coefficients can be interpreted as standard deviations; the sample is restricted to children from Jharkhand and the interior districts of Bihar; all regressions without district or year fixed effects include standalone treatment and post dummies but are not reported in the table; individual and household level controls include child's age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table D.4 – Effect of alcohol ban on reading and math scores of children in interior Bihar: alternative sample

	Ever Enrolled		All Children	
	Reading Score	Math Score	Reading Score	Math Score
Bihar Interior $\times$ Post	0.096*** (0.026)	0.124*** (0.027)	0.091*** (0.025)	0.118*** (0.026)
District FE	Yes	Yes	Yes	Yes
Survey Year FE	Yes	Yes	Yes	Yes
Birth Cohort FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	220,055	219,208	238,664	237,720
R-squared	0.433	0.437	0.457	0.459

Notes: The dependent variables are reading and math score which has been standardized with respect to the control group mean and standard deviation; the coefficients can be interpreted as standard deviations; the sample is restricted to children from Jharkhand and the interior districts of Bihar; individual and household level controls include child's age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table D.5 – Effect of alcohol ban on the age-standardized reading score of children in interior Bihar

	Standardized Reading Score				
	(1)	(2)	(3)	(4)	(5)
Bihar Interior $\times$ Post	0.094** (0.038)	0.089** (0.037)	0.094** (0.037)	0.093** (0.037)	0.097*** (0.035)
District FE	No	Yes	Yes	Yes	Yes
Survey Year FE	No	No	Yes	Yes	Yes
Birth Cohort FE	No	No	No	Yes	Yes
Controls	No	No	No	No	Yes
Observations	225,938	225,938	225,938	225,668	190,024
R-Squared	0.003	0.028	0.037	0.038	0.199

Notes: The dependent variable is the reading score, which has been standardized with respect to the control group mean and standard deviation of the particular age group of child; the coefficients can be interpreted as standard deviations; the sample is restricted to children from Jharkhand and the interior districts of Bihar; all regressions without district or year fixed effects include standalone treatment and post dummies but are not reported in the table; individual and household level controls include child's age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table D.6 – Effect of alcohol ban on the age-standardized math score of children in interior Bihar

	Standardized Math Score				
	(1)	(2)	(3)	(4)	(5)
Bihar Interior $\times$ Post	0.135*** (0.040)	0.134*** (0.040)	0.137*** (0.039)	0.135*** (0.039)	0.127*** (0.036)
District FE	No	Yes	Yes	Yes	Yes
Survey Year FE	No	No	Yes	Yes	Yes
Birth Cohort FE	No	No	No	Yes	Yes
Controls	No	No	No	No	Yes
Observations	224,814	224,814	224,814	224,547	189,232
R-Squared	0.007	0.026	0.050	0.050	0.213

Notes: The dependent variable is the math score, which has been standardized with respect to the control group mean and standard deviation of the particular age group of child; the coefficients can be interpreted as standard deviations; the sample is restricted to children from Jharkhand and the interior districts of Bihar; all regressions without district or year fixed effects include standalone treatment and post dummies but are not reported in the table; individual and household level controls include child's age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table D.7 – Effect of alcohol ban on the reading score of children in interior Bihar: with state-specific time trends

	Standardized Reading Score					
	(1)	(2)	(3)	(4)	(5)	(6)
Bihar Interior $\times$ Post	0.111*** (0.029)	0.105*** (0.029)	0.109*** (0.028)	0.074** (0.030)	0.092 (0.061)	0.097* (0.050)
District FE	No	Yes	Yes	Yes	Yes	Yes
Survey Year FE	No	No	Yes	Yes	Yes	Yes
Birth Cohort FE	No	No	No	Yes	Yes	Yes
State-Specific Time Trends	No	No	No	No	Yes	Yes
Controls	No	No	No	No	No	Yes
Observations	225,938	225,938	225,938	225,668	225,688	190,024
R-Squared	0.001	0.021	0.027	0.360	0.360	0.471

Notes: The dependent variable is the reading score, which has been standardized with respect to the control group mean and standard deviation; the coefficients can be interpreted as standard deviations; the sample is restricted to children from Jharkhand and the interior districts of Bihar; all regressions without district or year fixed effects include standalone treatment and post dummies but are not reported in the table; individual and household level controls include child's age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table D.8 – Effect of alcohol ban on the math score of children in interior Bihar: with state-specific time trends

	Standardized Math Score					
	(1)	(2)	(3)	(4)	(5)	(6)
Bihar Interior $\times$ Post	0.145*** (0.032)	0.142*** (0.032)	0.144*** (0.031)	0.109*** (0.032)	0.0698 (0.063)	0.076 (0.053)
District FE	No	Yes	Yes	Yes	Yes	Yes
Survey Year FE	No	No	Yes	Yes	Yes	Yes
Birth Cohort FE	No	No	No	Yes	Yes	Yes
State-Specific Time Trend	No	No	No	No	Yes	Yes
Controls	No	No	No	No	No	Yes
Observations	224,814	224,814	224,814	224,547	224,547	189,232
R-Squared	0.005	0.025	0.035	0.365	0.365	0.478

Notes: The dependent variable is the math score, which has been standardized with respect to the control group mean and standard deviation; the coefficients can be interpreted as standard deviations; the sample is restricted to children from Jharkhand and the interior districts of Bihar; all regressions without district or year fixed effects include standalone treatment and post dummies but are not reported in the table; the estimated coefficient in the last column (6) has a p-value of 0.160; individual and household level controls include child's age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table D.9 – Propensity score matching

Variable	Unmatched	Mean		%bias	%reduct bias	t-test	
	Matched	Treated	Control			t	p< t
Child Gender	U	0.5387	0.52536	2.6		0.06	0.000
	M	0.53837	0.52536	-0.6	78.6	-1.26	0.209
School Class	U	4.0587	4.1363	-3.5		-8.03	0.000
	M	4.058	4.044	0.6	82.1	1.40	0.162
Mother Literate	U	0.39553	0.536096	7.1		16.59	0.000
	M	0.39555	0.38344	2.5	65.0	5.60	0.000
Father Literate	U	0.64293	0.61686	5.4		12.55	0.000
	M	0.64295	0.63702	1.2	77.2	2.79	0.005
Household Size	U	7.3281	6.7318	18.1		42.25	0.000
	M	7.7323	7.2589	1.9	89.3	3.96	0.000
Household Toilet	U	0.30295	0.1675	32.3		75.68	0.000
	M	0.30294	0.27083	7.7	76.3	16.02	0.000
Household Type (Semi-Pucca)	U	0.32382	0.17058	36.1		84.47	0.000
	M	0.32379	0.32734	-0.8	97.7	-1.71	0.087
Household Type (Pucca)	U	0.32825	0.15633	41.0		95.96	0.000
	M	0.32826	0.32349	1.1	97.2	2.29	0.022
Child Age	U	9.7201	9.7099	0.4		0.85	0.393
	M	9.72	9.7063	0.5	-34.8	1.12	0.263

Notes: The treated and non-treated children have been matched using kernel matching algorithm with common support; probit model has been used to generate propensity scores; U and M refer to the unmatched and matched samples, respectively.

Table D.10 – Effect of alcohol ban on reading score of children in interior Bihar: propensity score matching

	Standardized Reading Score				
	(1)	(2)	(3)	(4)	(5)
Bihar Interior $\times$ Post	0.073** (0.032)	0.071** (0.029)	0.079*** (0.029)	0.057* (0.031)	0.061** (0.028)
District FE	No	Yes	Yes	Yes	Yes
Survey Year FE	No	No	Yes	Yes	Yes
Birth Cohort FE	No	No	No	Yes	Yes
Controls	No	No	No	No	Yes
Observations	190,175	190,175	190,175	190,019	190,019
R-Squared	0.001	0.018	0.023	0.0355	0.466

Notes: The dependent variable is the reading score, which has been standardized with respect to the control group mean and standard deviation; the coefficients can be interpreted as standard deviations; the sample is restricted to children from Jharkhand and the interior districts of Bihar; all regressions without district or year fixed effects include standalone treatment and post dummies but are not reported in the table; individual and household level controls include child's age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

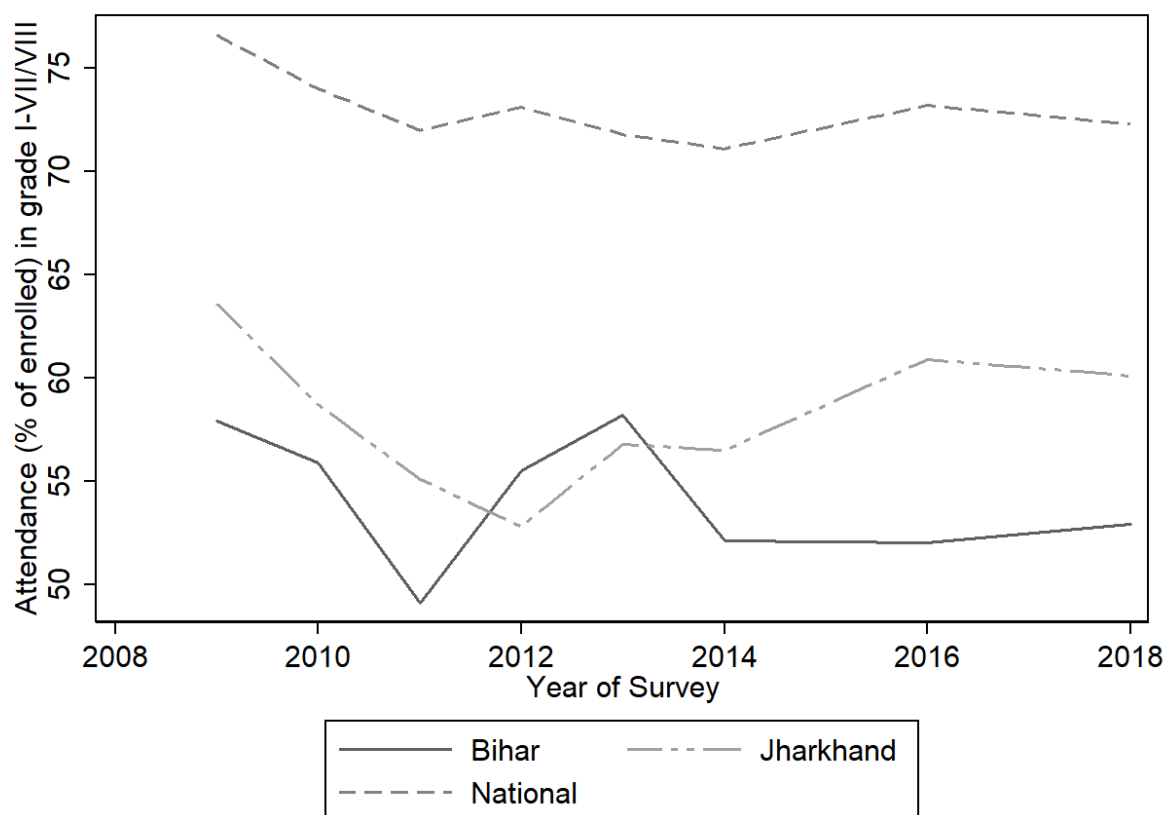
Table D.11 – Effect of alcohol ban on math score of children in interior Bihar: propensity score matching

	Standardized Math Score				
	(1)	(2)	(3)	(4)	(5)
Bihar Interior $\times$ Post	0.111*** (0.034)	0.109*** (0.033)	0.114*** (0.032)	0.092*** (0.032)	0.093*** (0.029)
District FE	No	Yes	Yes	Yes	Yes
Survey Year FE	No	No	Yes	Yes	Yes
Birth Cohort FE	No	No	No	Yes	Yes
Controls	No	No	No	No	Yes
Observations	189,382	189,382	189,382	189,227	189,227
R-Squared	0.002	0.021	0.029	0.361	0.474

Notes: The dependent variable is the math score, which has been standardized with respect to the control group mean and standard deviation; the coefficients can be interpreted as standard deviations; the sample is restricted to children from Jharkhand and the interior districts of Bihar; all regressions without district or year fixed effects include standalone treatment and post dummies but are not reported in the table; individual and household level controls include child's age, gender, grade, parental education status, family size, socioeconomic status, the availability of toilet inside the house; data source is the ASER household surveys from 2009-14, 2016 and 2018; standard errors in parentheses clustered at district level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

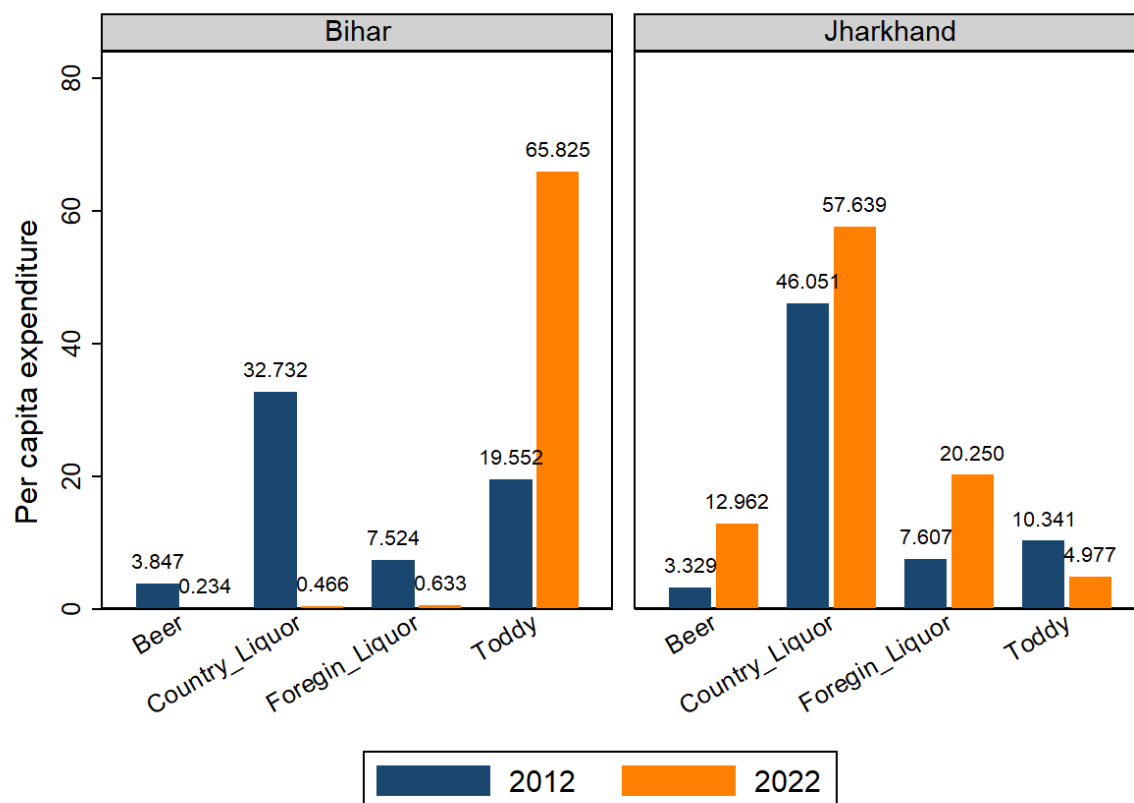
Figures

Figure D.1: Trends of school attendance in Bihar and Jharkhand over the years



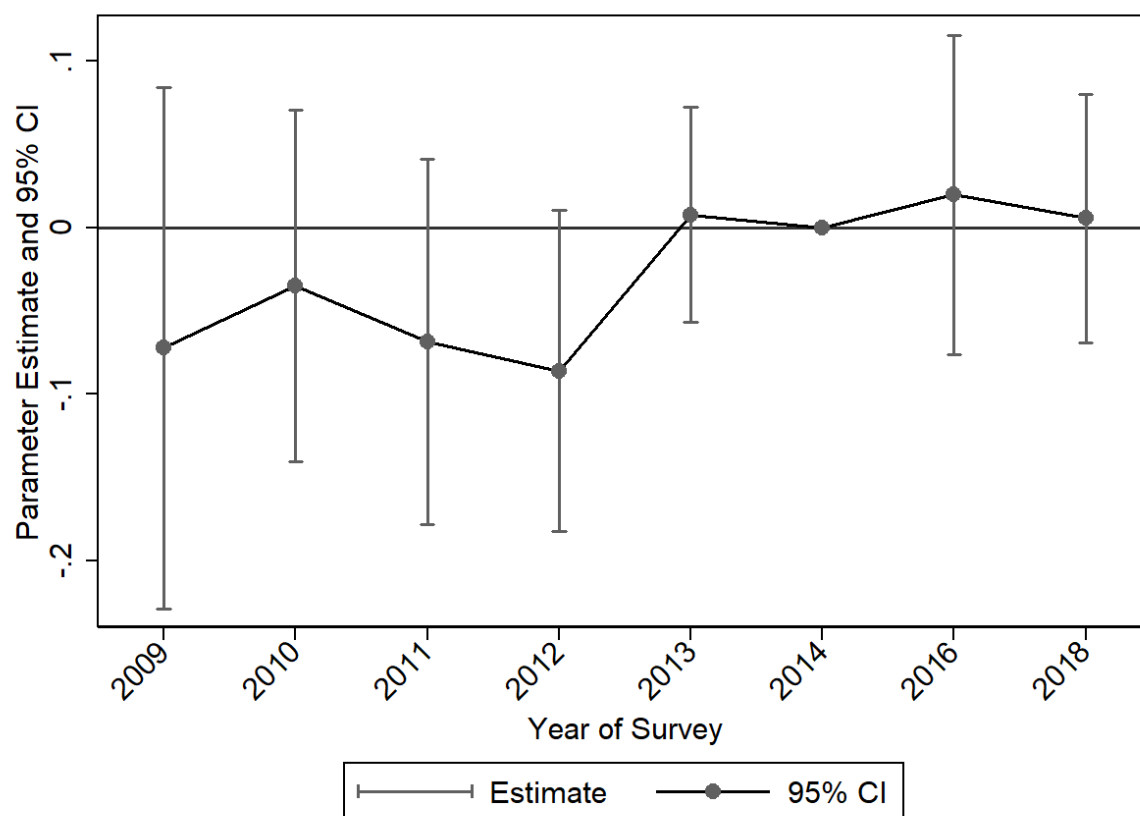
Notes: Data source is the periodic reports of ASER household surveys from 2009 - 2018.

Figure D.2: Real per capita consumption expenditure over different types of alcohol in Bihar vs Jharkhand



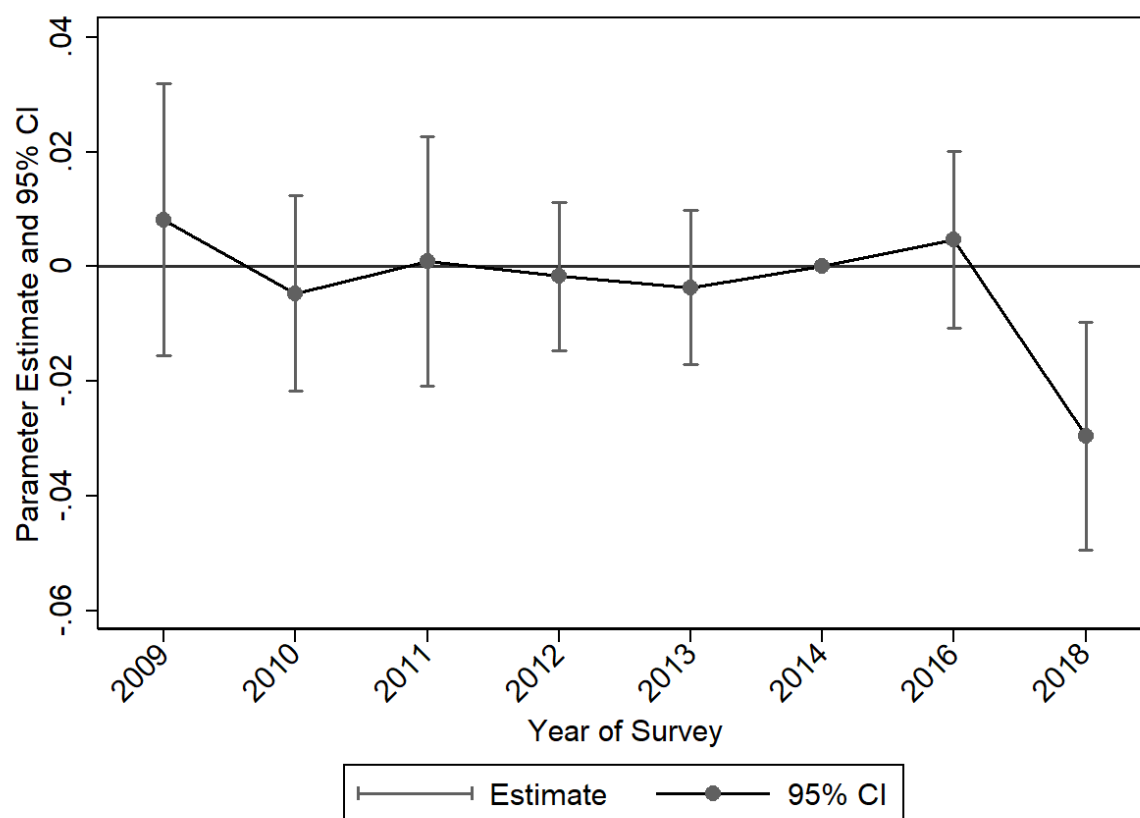
Notes: Data is extracted from the 68th and 79th rounds of NSSO consumption expenditure surveys conducted in 2011-12 and 2022-23, respectively; country liquor is Indian-made foreign liquor, foreign liquor comprises of different imported brands like whiskey and beer and toddy are locally brewed from fermented grains or coconut palm juice.

Figure D.3: Effect of alcohol ban on the completed years of schooling of children in interior Bihar: event study estimates



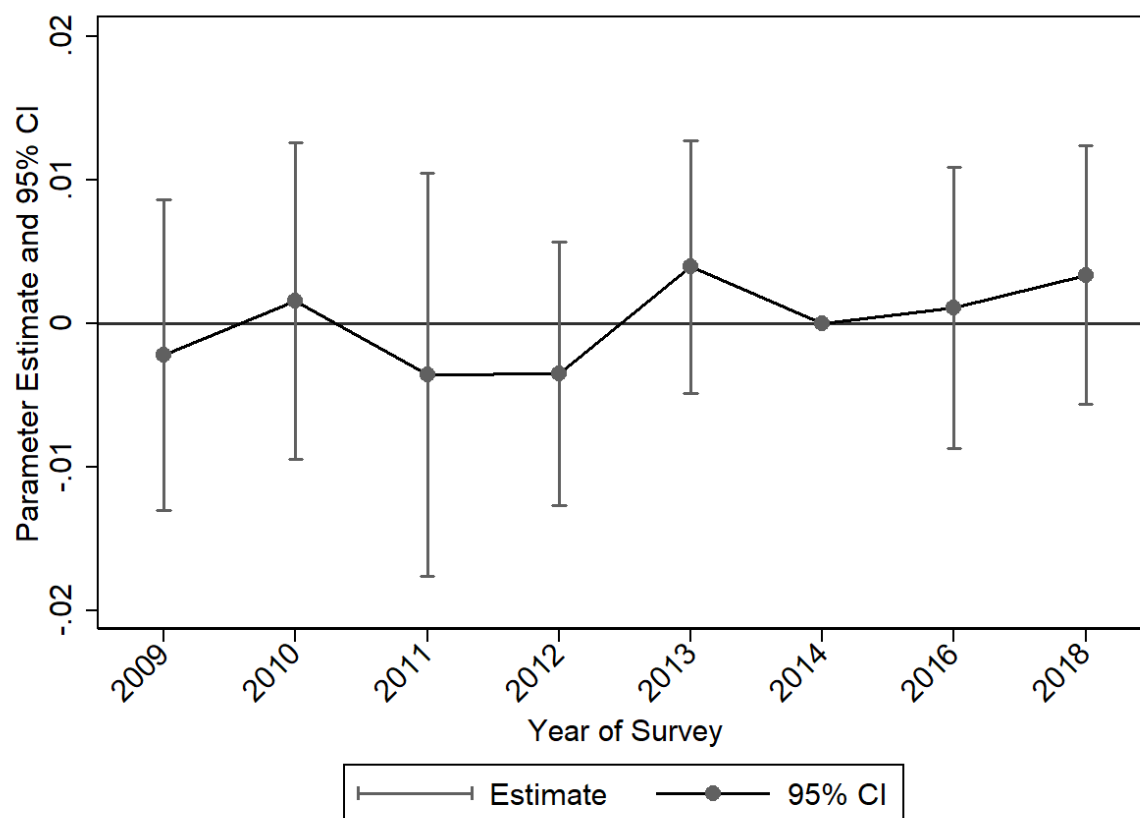
Notes: Wild clustered bootstrapped p-values for the two post periods, 2016 and 2018, are 0.698 and 0.783, respectively.

Figure D.4: Effect of alcohol ban on the enrollment (current) status of children in interior Bihar: event study estimates



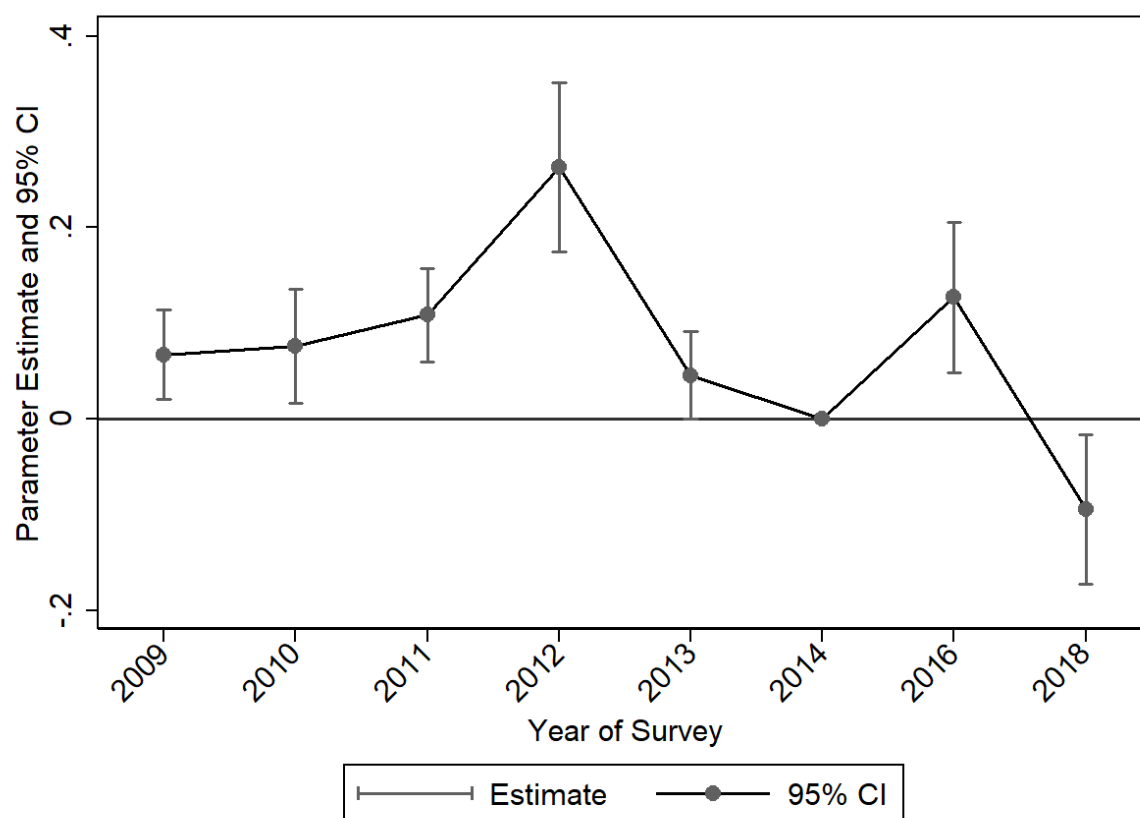
Notes: Wild clustered bootstrapped p-values for the two post periods, 2016 and 2018, are 0.560 and 0.005, respectively.

Figure D.5: Effect of alcohol ban on the dropout status of children in interior Bihar: event study estimates



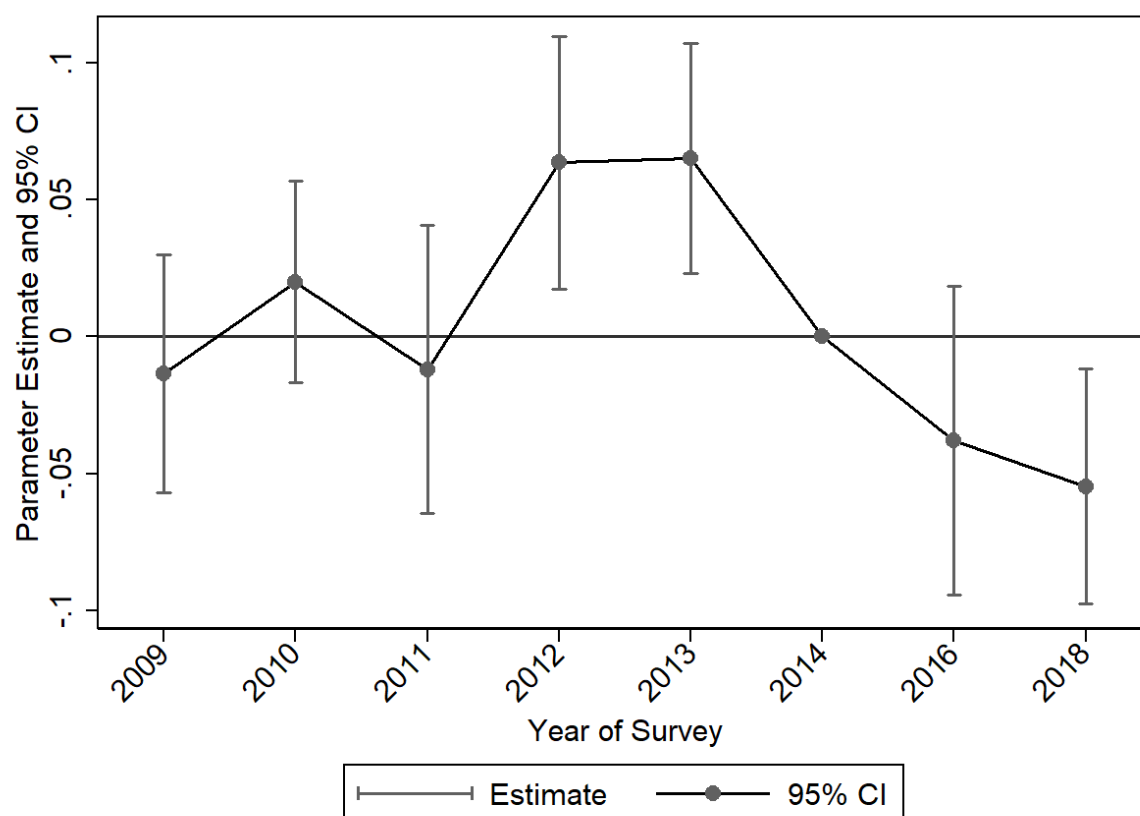
Notes: Wild clustered bootstrapped p-values for the two post periods, 2016 and 2018, are 0.82 and 0.461, respectively.

Figure D.6: Effect of alcohol ban on teachers' attendance in interior Bihar: event study estimates



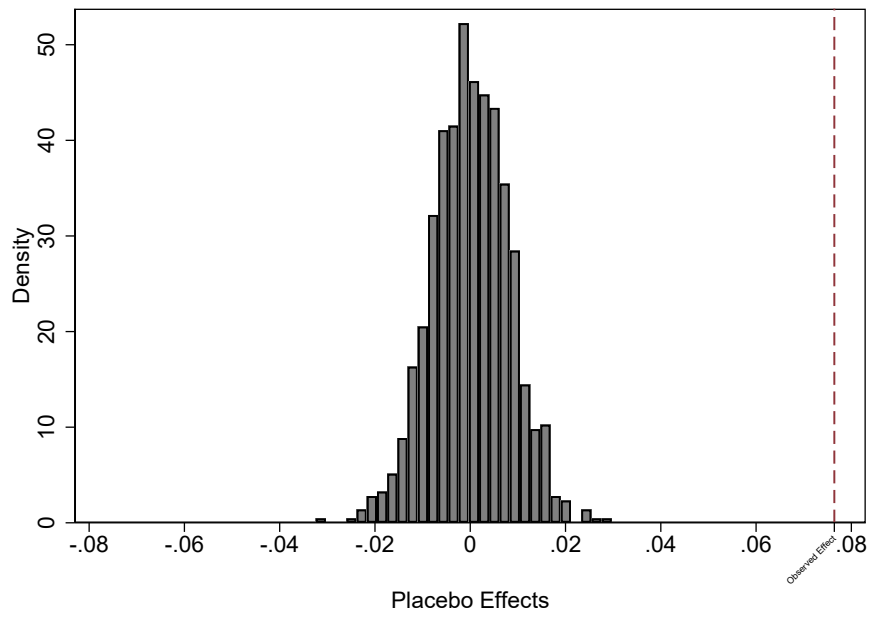
Notes: Wild clustered bootstrapped p-values for the two post periods, 2016 and 2018, are 0.000 and 0.016, respectively.

Figure D.7: Effect of alcohol ban on student attendance in interior Bihar: event study estimates

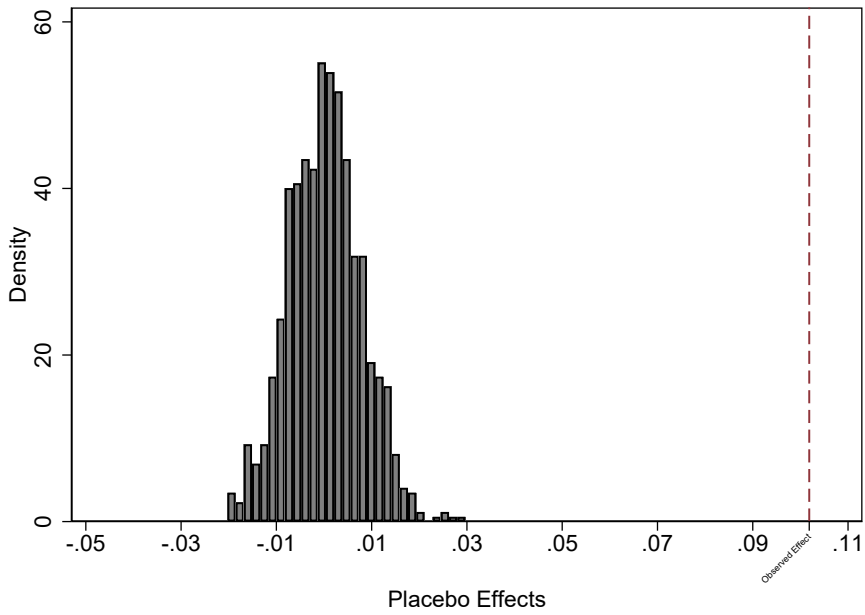


Notes: Wild clustered bootstrapped p-values for the two post periods, 2016 and 2018, are 0.168 and 0.018, respectively.

Figure D.8: Exact randomization (inference) test: kernel density plots



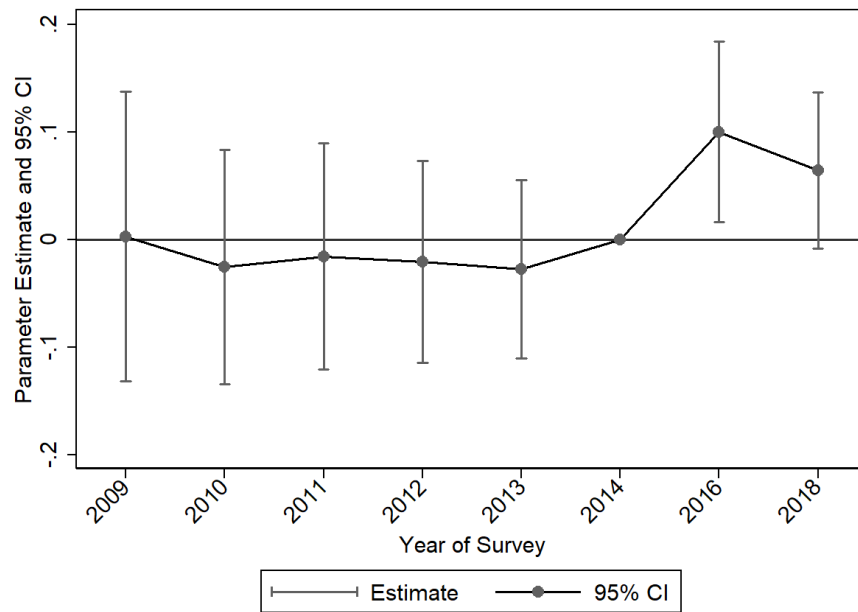
(a) Reading score



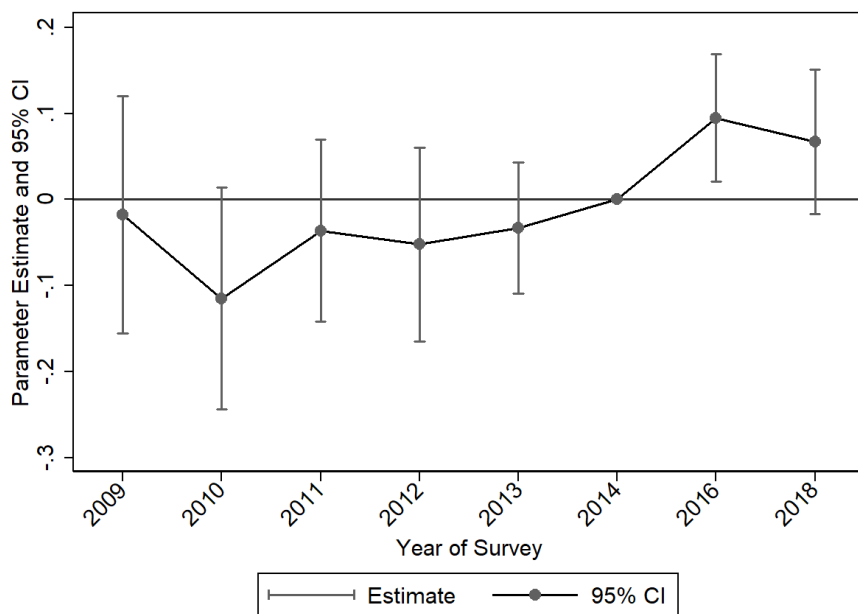
(b) Math score

Notes: Randomization tests are based on our main specification; we start by randomizing the treatment status and based on random assignment to treatment, we generate the interaction term for our DID model; we estimate our preferred specification and simulate 1,000 times; we collect these 1,000 coefficients and plot the distribution in this Figure; the vertical dashed line shows our original estimates for reference.

Figure D.9: Effect of alcohol ban on the reading and math score of children in interior Bihar (ever enrolled children): event study estimates



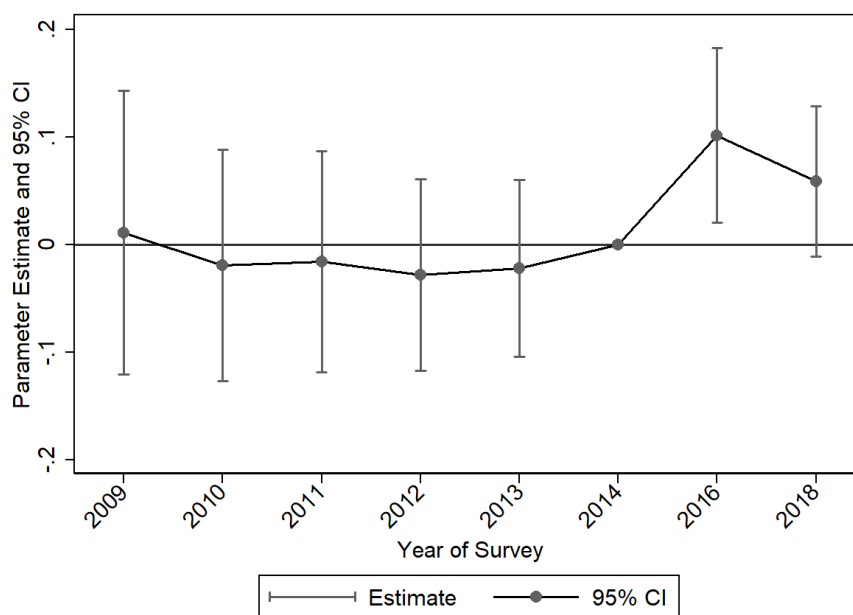
(a) Reading score



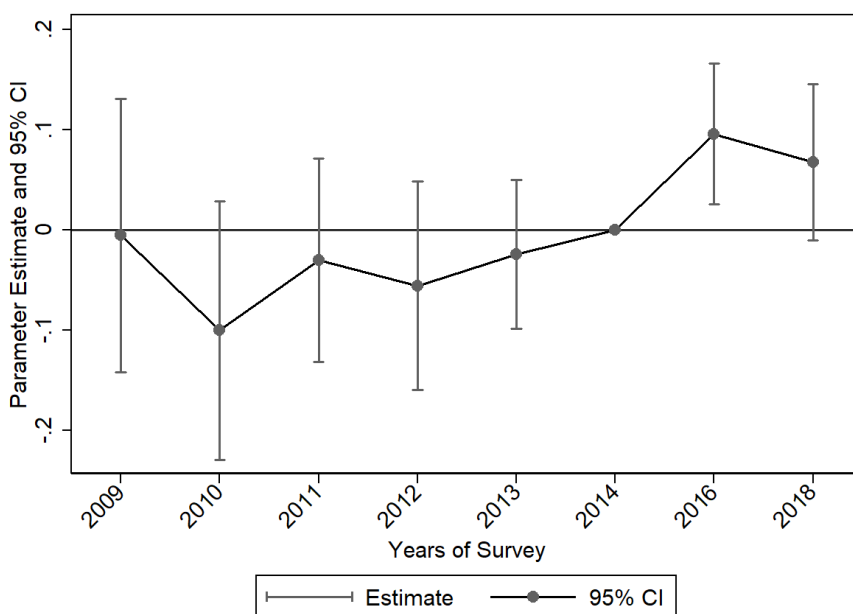
(b) Math score

Notes: Wild clustered bootstrapped p-values for the two post periods, 2016 and 2018, are 0.015 and 0.095 for reading and 0.018 and 0.135 for math scores, respectively.

Figure D.10: Effect of alcohol ban on the reading score of children in interior Bihar (all children irrespective of enrollment status): event study estimates



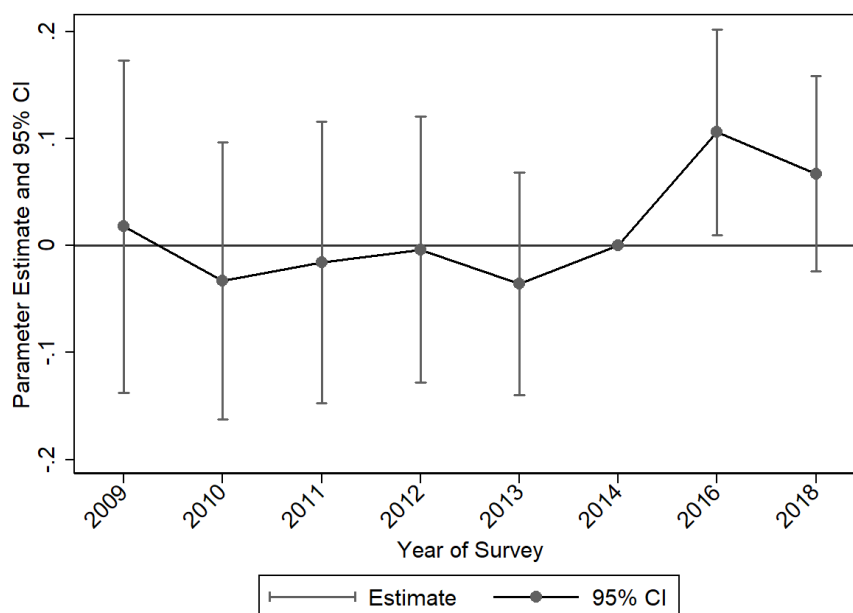
(a) Reading score



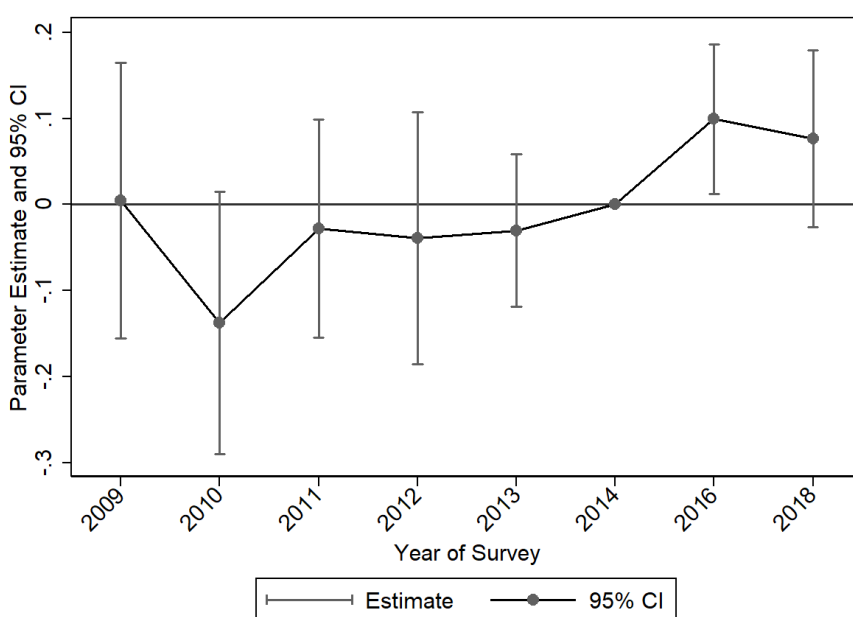
(b) Math score

Notes: Wild clustered bootstrapped p-values for the two post periods, 2016 and 2018, are 0.015 and 0.117 for reading and 0.018 and 0.095 for math scores, respectively.

Figure D.11: Effect of alcohol ban on the reading and math score of children in interior Bihar (control group-age standardized): event study estimates



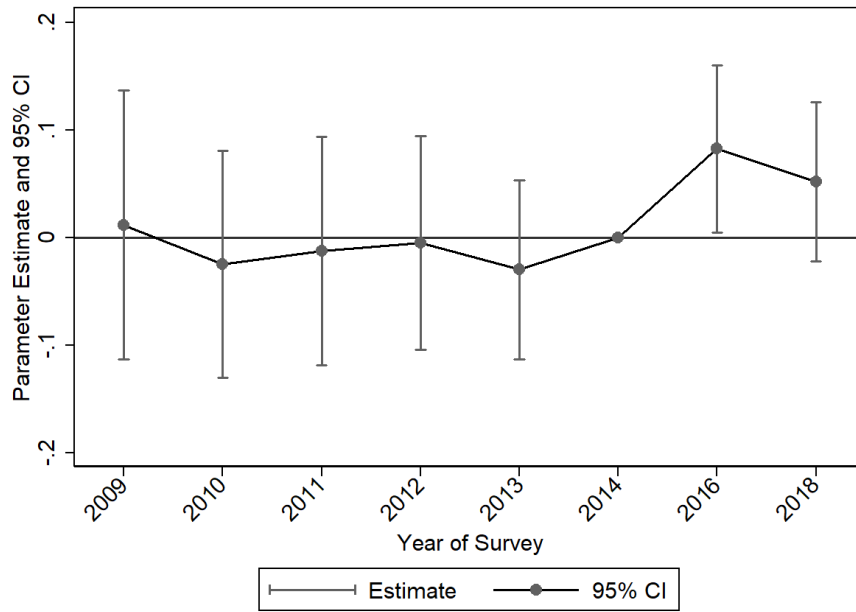
(a) Reading score



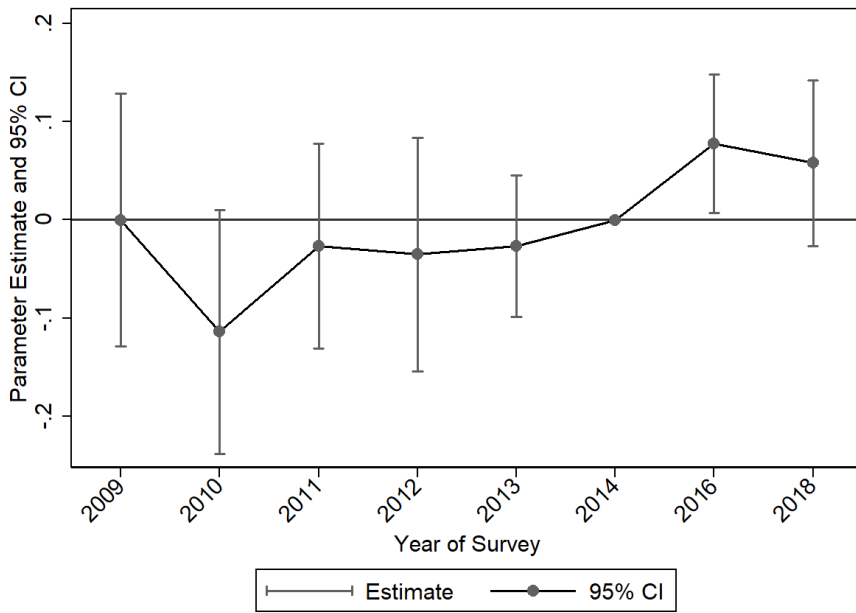
(b) Math score

Notes: Wild clustered bootstrapped p-values for the two post periods, 2016 and 2018, are 0.032 and 0.160 for reading and 0.021 and 0.152 for math score, respectively.

Figure D.12: Effect of alcohol ban on the reading score of children in interior Bihar (with state-specific time trends): event study estimates



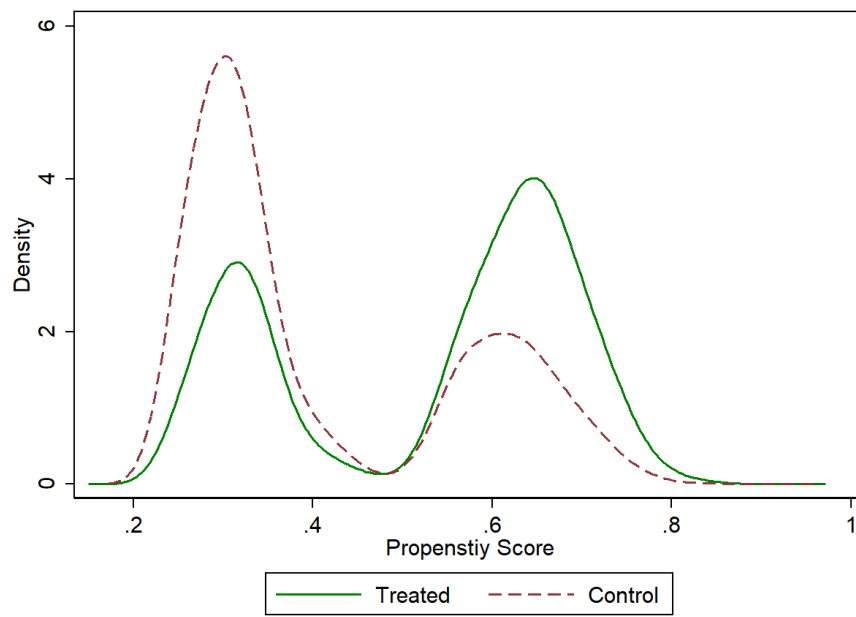
(a) Reading score



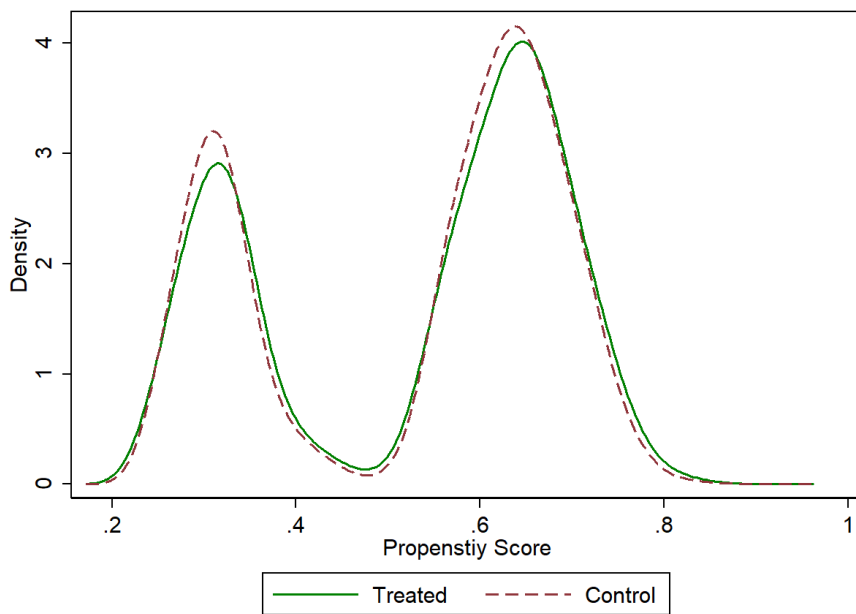
(b) Math score

Notes: Wild clustered bootstrapped p-values for the two post periods, 2016 and 2018, are 0.041 and 0.171 for reading and 0.0340 and 0.212 for math scores, respectively.

Figure D.13: PSM density plots



(a) Before matching



(b) After matching